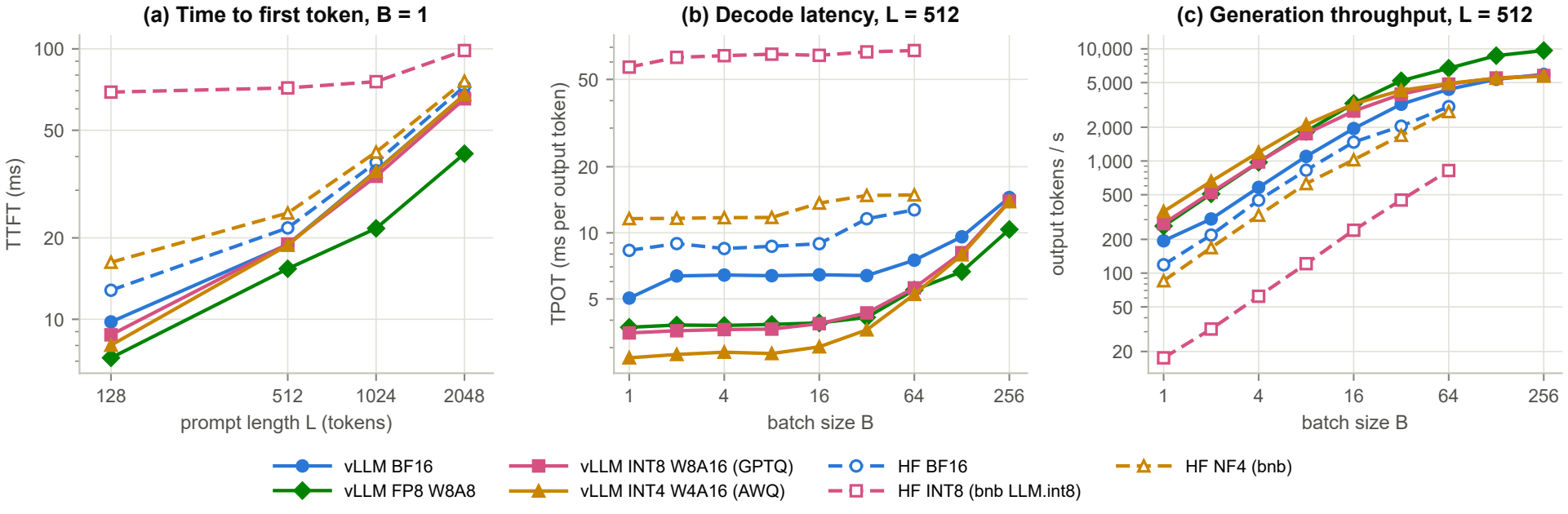

(a) Time to first token, B = 1
TTFT (ms)
100
50
20
10
128
512
1024
2048
prompt length L (tokens)
(b) Decode latency, L = 512
TPOT (ms per output token)
50
20
10
5
1
4
16
64
256
batch size B
(c) Generation throughput, L = 512
output tokens / s
10,000
5,000
2,000
1,000
500
200
100
50
20
1
4
16
64
256
batch size B
vLLM BF16
vLLM FP8 W8A8
vLLM INT8 W8A16 (GPTQ)
vLLM INT4 W4A16 (AWQ)
HF BF16
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)